

Presentation Script — Mid-Review

Deepfake Detection: XceptionNet vs. CNN-ViT Hybrid on FaceForensics++ · 23AID205 · Target: under 15:00, budgeted at 14:15

#	Presenter	Slides	Starts	Budget
1	Vipin Sudhakar	1–3	0:00	3:15
2	Rithvik Arulprakash	4–5	3:15	3:00
3	Venugopalan Gangadharan	6–8	6:15	4:15
4	Harshith Kv	9–11	10:30	3:45

Vipin Sudhakar

Starts at 0:00 — 3:15 budget

Slide 1 Title

0:30

- Good [morning/afternoon]. We're presenting our AIML course project for 23AID205: Deepfake Detection -- comparing XceptionNet against a CNN-ViT hybrid on FaceForensics++.
- In the next 15 minutes we'll cover our approach, three experiments' worth of real results, an interpretability analysis, and a live working demo.
- I'm Vipin -- I'll open with the motivation and our methodology, then hand off to my teammates.

Slide 2 Motivation, Problem & Objectives

1:30

- Deepfakes have moved from a research curiosity to a real misinformation and fraud vector.
- The problem: published detectors usually report one headline accuracy number, on the same manipulation type and compression level they trained on. That number is misleading -- a detector at 99% in a paper can drop to ~57% AUC, worse than random, the moment the test distribution shifts. And that failure is invisible until deployment.
- So our objectives were threefold: fairly compare XceptionNet against a CNN-ViT hybrid -- does adding global self-attention actually help generalization? Measure not just accuracy but full cross-manipulation generalization and compression robustness. And explain **why** generalization succeeds or fails, via Grad-CAM, rather than just reporting a number.

Slide 3 Methodology & Pipeline

1:15

- Our pipeline in six steps: raw FF++ video, MTCNN face extraction, official video-level splits, training, evaluation, and analysis.
- One thing worth flagging: we used the **official** video-level train/val/test splits throughout -- not random frame-level splits, which would leak the same identity into both train and test and quietly inflate accuracy.

Handoff: "Rithvik will now walk you through our dataset and the two architectures we compared."

Slide 4 Dataset, Preprocessing & EDA**1:30**

- FaceForensics++: 5,000 videos total -- 1,000 real YouTube interviews plus 4,000 manipulated, 1,000 each across four methods: Deepfakes, Face2Face, FaceSwap, and NeuralTextures, at c23 compression.
- Preprocessing: MTCNN face detection on GPU, 20 frames per video, a 1.3x crop margin to capture blending-boundary artifacts, giving us roughly 99,987 face crops in total.
- Before committing to full preprocessing, we ran an IQR outlier check on frame count and file size -- flagged 502 of 5,000 videos, about 10%, so data quality was verified upfront, not assumed.
- One honest caveat we'll return to: official raw and heavily-compressed data access from FaceForensics++ was still pending at review time -- we'll show how we handled that later.

Slide 5 Models: XceptionNet vs. CNN-ViT Hybrid**1:30**

- XceptionNet is our baseline -- a standard depthwise-separable CNN, ImageNet-pretrained, purely convolutional with only local receptive fields.
- The CNN-ViT hybrid pairs a ResNet-50 feature extractor with a 6-layer, 8-head Transformer encoder. The question we're testing: does global self-attention let the model relate distant regions -- like blending inconsistencies -- that a local CNN structurally can't see jointly?
- Critically, both models share the exact same optimizer, schedule, batch size, and data. Architecture is the only variable, by design -- so any difference in results is attributable to that choice alone.

Handoff: "Venu will now take you through what we actually found across our three experiments."

Slide 6 Baseline Comparison**1:15**

- Both models trained and tested on all four methods together -- the same-distribution, headline number most papers stop at.
- XceptionNet: 97.29% video accuracy, 0.9958 AUC. CNN-ViT hybrid: 94.71% video accuracy, 0.9836 AUC.
- XceptionNet leads on every single metric here -- our first sign that the hybrid's added complexity isn't paying off on this task.

Slide 7 Cross-Manipulation Generalization**1:45**

- This is the actual novel question we set out to answer: train on ONE manipulation method, test on all four -- a full 4x4 matrix, per model.
- [Point to heatmap] The diagonal -- same-manipulation -- stays high across the board for both models. Off-diagonal is where it falls apart.
- Our generalization gap: XceptionNet 0.398, CNN-ViT hybrid 0.421. XceptionNet actually generalizes **better** than the attention-based hybrid -- the opposite of our original hypothesis. We're reporting that honestly as a

genuine negative result, not hiding it.

- Worst case for both: a model trained only on Deepfakes, tested on FaceSwap -- AUC of 0.256 and 0.198. That's worse than random guessing.

Slide 8 Compression Robustness

1:15

- Official c0 (raw) and c40 (heavy compression) access from FaceForensics++ was still pending, and no c40 mirror exists on Kaggle -- so rather than skip this experiment, we built a documented workaround.
- We locally re-transcoded just our held-out test videos to a harsher compression level with ffmpeg -- we call it 'proxy-c40.' We're upfront that this double-compresses rather than compressing once from raw, so it's an approximation -- used only for evaluation, never for training.
- Result: both models drop from ~0.99 AUC to roughly 0.80-0.82 under heavier compression. The hybrid degrades slightly less -- a 0.166 drop versus 0.194 for XceptionNet.

*Handoff: "Harshith will now show you *why* this generalization gap happens, and our live demo."*

Harshith Kv

Starts at 10:30 — 3:45 budget

Slide 9 Grad-CAM: Why Generalization Fails

1:30

- We wanted to explain the gap, not just report it. This is Grad-CAM on the same video, same Deepfakes-only checkpoint, tested on two different manipulation methods.
- Same-manipulation: attention locks tightly onto the central face -- nose, mouth, eyes -- exactly where Deepfakes' blending artifacts show up. Confident, and interpretably correct.
- Cross-manipulation, same video, switched to FaceSwap: XceptionNet's attention collapses to a random spot off the face entirely. The hybrid's attention goes diffuse across the whole frame, including the background.
- So neither model is merely 'uncertain' on unseen manipulations -- each confidently learned a narrow signal, with nothing sensible to fall back on once that exact signal is gone.

Slide 10 Live Demo

1:15

- [Switch to the live demo] Upload a held-out test video, get a real/fake probability from either trained model -- the pipeline mirrors training exactly: face crop, normalize, per-frame score, video-level average.
- We verified this against a curated set of test-split videos never seen in training -- correctly and confidently classified.
- One honest moment worth sharing: we also tested it against a random deepfake-claim image circulating online, outside FaceForensics++ entirely. Both models were confidently wrong -- which is exactly what our generalization-gap numbers predicted, not a contradiction of them.

Slide 11 Conclusion & Closing

1:00

- To close: XceptionNet outperforms the CNN-ViT hybrid on every metric we measured, contradicting our initial hypothesis that attention would help.
- Both architectures share the same underlying weakness -- narrow, method-specific artifacts rather than a general 'this is manipulated' signal -- which we confirmed visually, not just numerically.

- Future work: official c0/c40 once FaceForensics++ approval comes through, multi-seed runs for statistical confidence, and extending Grad-CAM to a genuinely-uncertain case rather than only the worse-than-random one.
- Thank you -- happy to take questions.